

TwinVerse Inspect AI – Complete Development Blueprint

Vision

TwinVerse Inspect AI is an AI-powered Infrastructure Intelligence Platform that analyzes images and videos from drones, CCTV cameras, robots, and smartphones to detect defects such as cracks, corrosion, surface damage, equipment faults, and structural anomalies. The platform produces actionable maintenance insights and visualizes infrastructure health through a digital twin.

Hackathon Goal

Build a working MVP that demonstrates image/video ingestion, AI defect detection, severity scoring, reporting, and digital-twin visualization. Focus on explainability and practicality rather than trying to build a perfect enterprise product.

System Architecture

Data Sources → Upload Service → AI Inference Engine → Defect Database → Severity Assessment → Digital Twin Viewer → Dashboard → Reports.

Recommended Tech Stack

Frontend: React + Next.js Backend: FastAPI (Python) Database: PostgreSQL Storage: MinIO or S3-compatible storage AI: YOLOv11 / RT-DETR + OpenCV 3D Visualization: Three.js Digital Twin: Gaussian Splatting / NeRF (future) Simulation: Gazebo Harmonic and NVIDIA Isaac Sim Deployment: Docker + GitHub Actions

Development Roadmap

Phase 1: Upload images and videos. Phase 2: AI crack/corrosion detection. Phase 3: Severity scoring. Phase 4: Dashboard and reports. Phase 5: Digital twin visualization. Phase 6: Drone integration. Phase 7: Predictive maintenance.

Beginner-Friendly Build Order

Week 1: Learn Git, GitHub, Python, FastAPI. Week 2: Build upload API. Week 3: Train/download defect detector. Week 4: Build dashboard. Week 5: Add digital twin. Week 6: Demo preparation.

AI Models

Crack Detection, Corrosion Detection, Surface Damage Detection, Missing Component Detection, Thermal Anomaly Detection (future).

Digital Twin

Start with a simple 3D asset viewer and defect markers. Full digital twins can be added later using photogrammetry or NeRF pipelines.

Simulation

Use Gazebo for robotics simulation and Isaac Sim for synthetic dataset generation and future autonomous inspection workflows.

Security Requirements

JWT authentication, HTTPS, RBAC, encrypted secrets, GitHub secret scanning, input validation, audit logging, Docker security hardening.

GitHub Instructions for Claude

Create a professional repository structure. Generate README.md. Create CONTRIBUTING.md. Add Docker support. Configure GitHub Actions CI/CD. Write beginner-friendly documentation. Use clean commits and branches. Create issue templates and project board.

README Requirements

Project overview, architecture diagram, setup guide, Docker guide, API guide, AI training guide, deployment guide, roadmap, screenshots.

Judging Advantages

Explainability, real-world applicability, infrastructure safety, predictive maintenance, cloud readiness, digital twin visualization, scalable architecture.

Can You Build It?

Yes. A complete enterprise platform would take a team and months. However, a strong hackathon MVP is absolutely achievable by focusing on image ingestion, AI detection, severity scoring, reporting, and a simple digital twin viewer.

Claude Master Prompt

You are the lead architect and senior software engineer for TwinVerse Inspect AI. Build this project from scratch for a beginner developer. Explain every step, generate production-quality code, maintain documentation, push changes to GitHub, write tests, add security best practices, keep architecture simple, and never assume prior experience. Every feature must include explanations, folder structure, commands, and verification steps.

End of Proposal